STRAViT: A Spatial–Temporal Feature-Reshaped Agent Vision Transformer for EEG-Based Emotion Recognition

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***Abstract*—Emotion recognition from electroencephalogram (EEG) signals has attracted increasing interest due to its applications in affective computing and brain–computer interaction. However, effectively integrating spatial–temporal features while attending to emotionally relevant brain regions remains a significant challenge. In this article, we propose a spatial–temporal feature-reshaped agent vision transformer (STRAViT) network, which fuses differential entropy (DE) and functional connectivity matrices through a dual-branch attention-based feature reshaping module (FRM), enabling refined feature integration. In addition, we introduce an agent vision transformer (AViT) that utilizes learnable agent tokens to capture global–local dependencies within EEG representations efficiently. Extensive experiments conducted on the SEED and SEED-IV datasets demonstrate that the STRAViT achieves classification accuracies of 98.7% and 87.2% on SEED, 95.5% and 72.5% on SEED-IV, for discrete emotion recognition, under subject-dependent and subject-independent strategies. The proposed network not only effectively integrates spatial–temporal features of EEG signals but also enhances the modeling of inter-regional dependencies through agent attention. Comprehensive tests confirm that the STRAViT yields excellent performance on emotion recognition tasks.**

***Index Terms*—Agent vision transformer (AViT), attention, electroencephalogram (EEG) emotion recognition, feature fusion.**

I. Introduction

EMOTION recognition has increasingly become a research hotspot in the field of artificial intelligence. Emotion, as an intricate interplay of physiological and psychological processes, is fundamental to human cognition, behavior modulation, and interpersonal communication [1]. Achieving accurate recognition of emotional state not only helps improve the intelligence level of human–computer interaction (HCI) systems [2], but also provides essential support for mental health, intelligent monitoring, and other fields [3]. Currently, emotion recognition approaches are generally divided into two main categories: those utilizing nonphysiological signals and those based on physiological data [4]. Nonphysiological methods commonly depend on external behavioral indicators, such as facial expressions [5] and speech [6], which are relatively easy to acquire. However, individuals are capable of deliberately controlling nonphysiological cues, like facial expressions and vocal tones, to mask their genuine emotions, especially in social contexts. This limits the practical reliability of these approaches in realistic application scenarios. In contrast, physiological signals have drawn more attention from researchers because they can more accurately depict inherent changes in feelings. Among these, electroencephalogram (EEG) has gained increasing attention in emotion identification research owing to its noninvasive nature, real-time accessibility, low cost, as well as its high temporal resolution, and difficulty in being concealed [7].

Emotion modeling is generally described from two perspectives [8]: discrete emotion representation and dimensional emotion representation. The former categorizes emotions into a finite set of distinct emotional states (e.g., joy, sadness, surprise, fear, anger, and disgust) [9], whereas the latter characterizes emotions along continuous dimensions such as valence and arousal, reflecting emotional polarity and activation level, respectively [10]. In the context of EEG-based emotion recognition, discrete emotion classification has been widely adopted, as distinct emotional states are often associated with discriminative neural activation patterns and inter-regional connectivity structures. A growing body of studies has shown that explicitly modeling emotion-related brain characteristics—such as hemispheric asymmetry, inter-regional information propagation, and activation discrepancies—can significantly enhance the discrimination of discrete emotional states in EEG signals [11], [12]. For instance, bihemispheric discrepancy model (BiHDM) has been proposed to capture asymmetric emotional responses between the left and right hemispheres, leading to improved performance in discrete emotion classification [13]. Other studies have demonstrated that incorporating channel-wise selection strategies, brain activation differences, and connectivity-aware representations further strengthens

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the extraction of emotion-discriminative features from EEG recordings [14], [15], [16]. These findings collectively indicate that structured neural patterns and region-level interactions provide an effective basis for discrete EEG emotion recognition.

Over the past few years, EEG-based emotion recognition has garnered significant research interest, driving the adoption of a wide range of machine learning techniques. Traditional approaches, including support vector machines (SVMs) [17] and k-nearest neighbors (KNNs) [18], have been commonly used in this field. With the rapid development of deep learning, advanced models such as convolutional neural networks (CNNs) [19] and long short-term memory (LSTM) networks [20] have exhibited promising performance in EEG-based emotion analysis.

Despite progress in emotion recognition, several challenges persist in EEG-based approaches. In particular, most existing methods rely on single-type features, which limits the exploration of the multidimensional characteristics of EEG signals [21]. The joint modeling of spatial–temporal information, capturing both intrachannel and interchannel relationships, has not been fully explored. Then, although some works attempt to incorporate various types of features, they often lack an effective fusion mechanism. In addition, although transformers have been widely applied in EEG analysis due to their strong modeling capabilities [22], how to effectively design methods that align with the specific properties of EEG signals remains an open research question. Therefore, efficiently integrating diverse features and improving generalization capability have become key directions for advancing EEG-based emotion recognition.

To overcome the limitations discussed above, we introduce a spatial–temporal feature-reshaped agent vision transformer (STRAViT) network for EEG-based emotion recognition. Specifically, we design a feature reshaping module (FRM) that fuses differential entropy (DE) features with functional connectivity matrices via a dual-branch attention mechanism. This mechanism applies weighted modeling along channel and spatial dimensions to enhance salient regions while suppressing less informative ones. A reverse attention branch is further incorporated to recover potentially overlooked but valuable features, achieving a balance between saliency enhancement and information completeness. In addition, we integrate an agent attention [23] into the vision transformer (ViT) [24] for EEG-based emotion recognition, which replaces traditional token-to-token self-attention with a small set of learnable agent tokens. We investigate its potential for efficient feature aggregation and attention guidance, aiming to address the challenges of redundant information and weak spatial dependency in EEG signals. This network, designed for EEG emotion recognition, effectively focuses on emotion-relevant regions and significantly enhances generalization across subjects, underscoring its practical value and novelty.

The key contributions of this work are outlined below.

1) We propose a STRAViT model for EEG emotion recognition, which incorporates dual inputs from DE and functional connectivity matrices to comprehensively capture both local energy distributions and cross-channel dependencies in EEG signals. This unified architecture effectively models both local and global characteristics of EEG signals, contributing to better emotion recognition accuracy.

2) The FRM is constructed with a dual-branch attention mechanism for feature fusion and refinement. The main branch emphasizes salient regions, while the auxiliary branch employs reverse attention to recover complementary and previously overlooked information, effectively reducing redundancy while preserving critical details to improve feature completeness.

3) We design an agent ViT (AViT) tailored for EEG-based emotion recognition, with a focus on discrete emotion recognition, which significantly improves classification performance by introducing agent tokens to capture global–local dependencies. The experimental evaluation verifies the superior capability and effectiveness of the proposed STRAViT framework.

II. Related Work

A. Feature Extraction Based on EEG

Current EEG-based emotion recognition research primarily focuses on feature extraction and the design of classification models. Most existing studies extract features using single-channel analysis. Cichocki et al. [25] compared DE and power spectral density (PSD), showing DE performs better. However, these studies have focused only on single-channel feature extraction, ignoring the spatial information of the EEG. To address this, Jia et al. [26] used graph convolutional networks to model local and global dependencies via adjacency matrices. Chen et al. [27] fused frequency-domain and connectivity features to improve cross-subject EEG emotion recognition. Li et al. [28] combined DE with functional connectivity to capture both temporal and spatial information. Guo et al. [29] introduced the FC-FAN model, leveraging EEG and functional connectivity data for richer and complementary feature learning. Fu et al. [30] proposed a linear–nonlinear feature reconstruction approach that jointly constructs and fuses multiple source EEG features from static and dynamic brain functional connectivity, thereby enhancing feature discriminability for emotion recognition. These studies demonstrate that multidimensional features provide more comprehensive EEG representations and enhance classification performance.

B. EEG Emotion Recognition Based on Deep Learning

Numerous deep learning-based models have been developed for EEG classification as deep learning techniques continue to evolve. Zheng and Lu [10] applied deep belief networks to EEG emotion recognition, leveraging DE to identify key frequency bands and channels. Song et al. [31] introduced a dynamical graph CNN (DGCNN) to capture the dynamic relationships among EEG channels. To leverage hemispheric information, Li et al. [32] introduced the bi-hemisphere domain adversarial neural network (BiDANN) and the BiHDM [13]. Recent studies have further advanced deep learning techniques, introducing innovations to classic models from various perspectives. Hou et al. [33] employed a residual feature pyramid network and replaced CNN with a

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space-to-depth structure to reduce computational complexity. Wang et al. [34] proposed a hybrid spatial–temporal feature fusion neural network (STFFNN) that integrates CNN-based spatial learning, temporal modeling, and Bi-LSTM-based fusion, achieving robust and discriminative EEG emotion recognition performance. Recent studies have introduced contrastive learning [35] and capsule network [36] into EEG-based emotion classification, expanding the range of effective modeling techniques. Dynamic graph attention networks have been employed to model non-Euclidean dependencies between EEG channels via multihead self-attention [37]. To address temporal variability, a temporal-difference minimizing network was introduced, leveraging maximum mean discrepancy and a multibranch convolutional recurrent architecture [38]. In addition, multiscale approaches such as DSSTNet have demonstrated effectiveness in extracting spatial, frequency, and temporal features simultaneously [39]. Attention mechanisms have attracted widespread interest due to their outstanding performance in deep learning [40], [41], [42]. They dynamically emphasize important features, thereby enhancing model representation, and have demonstrated strong potential in EEG-based emotion recognition.

C. EEG Emotion Recognition Based on Transformer

Transformer has attracted increasing attention in EEG-based emotion recognition due to their strong capacity for modeling global dependencies via self-attention. ViT, proposed by Google [24], extends the Transformer framework to visual domains by treating image patches as sequences. Inspired by this, several studies have applied Transformer and its variants to capture complex spatiotemporal patterns in EEG signals [43], [44], [45], [46], [47]. Song et al. [48] introduced a compact convolutional Transformer that unifies local and global information. Xu et al. [49] proposed spectral–spatial transformer blocks and temporal attention blocks to focus on informative EEG segments. Lu et al. [50] designed a Convolution Interactive Transformer to fuse local–global features. Su et al. [51] introduced the ViT module to enhance the extracted features' discriminative capability in EEG-based emotion recognition. A self-attention-enhanced MBConv module was proposed to capture both global and local EEG features [52]. Building on the ViT, researchers have made extensive improvements to the structure and mechanism of its core attention module to enhance adaptability to various data types. Han et al. [23] proposed agent attention, which balances representation capacity and computational cost. Given its ability to preserve global context modeling while offering significantly higher efficiency than conventional Softmax attention, we incorporate agent attention into EEG-based emotion recognition to enhance the model's sensitivity to emotion-related brain regions and improve overall feature representation.

D. EEG Emotion Recognition Based on Cross-Subject

Cross-subject EEG emotion recognition presents a crucial yet complex challenge, primarily due to considerable interindividual variability in emotional expression. The core difficulty lies in the distribution discrepancy of features across subjects. To alleviate this, transfer learning has been widely adopted, as it leverages knowledge from source domains to enhance performance on target domains with limited or no labeled samples [53], [54]. Among various transfer strategies, domain adaptation plays a central role by aligning the feature distributions between source and target subjects, thus improving the model's generalization to previously unseen individuals [55], [56]. Luo et al. [57] proposed a graph-based class-domain feature selection mechanism that evaluates feature relevance based on class-wise separability while accounting for domain-level variability. Zhong et al. [58] proposed an unsupervised domain adaptation method combined with pseudolabel propagation, which leverages global distribution alignment and reliable pseudolabel propagation to achieve robust crossdomain EEG emotion recognition performance. Additionally, Chen et al. [59] introduced an unsupervised EP transformation module based on VAE to reduce session-dependent EP differences. They further incorporated five segmental activation functions to facilitate the alignment of emotional patterns across subjects.

III. Methodology

To address the limitations in feature representation and the underutilization of information in EEG-based emotion recognition tasks, we propose a novel model named STRAViT. The overall architecture of STRAViT is illustrated in Fig. 1. The model mainly consists of three components: a feature extraction block including dual inputs from DE and functional connectivity matrices, FRM, and AViT.

A. Feature Extraction

1) DE Extraction: DE is chosen in our study for its proven effectiveness in capturing the complexity of EEG signals. DE has been extensively employed due to its strong ability to characterize emotion-related neural activities, and its effectiveness in EEG-based emotion recognition has been well validated in previous studies [21], [60]. Single-channel DE is adopted as it provides a compact and physiologically meaningful measure of band-specific EEG activity, and has demonstrated strong robustness and cross-subject consistency in emotion recognition tasks. To extract EEG features that contain both temporal and frequency information, we adopt a DE-based feature extraction method. Specifically, each raw EEG trial is initially divided into fixed-length segments (e.g., one second). To ensure a balanced number of samples per class during training, we perform resampling to obtain N EEG segments from each trial. Each segment is then bandpass filtered to isolate five frequency bands: $\delta(1-4Hz), \theta(4-8Hz), \alpha(8-14Hz), \beta(14-31Hz), \gamma(31-50Hz)$. For each frequency band, DE features are calculated using the following formula:

$$DE = -\int_{-\infty}^{\infty} P(x) \log(P(x))\, dx \tag{1}$$

where x is a random variable representing the amplitude of the EEG signal, and $P(x)$ represents the probability density

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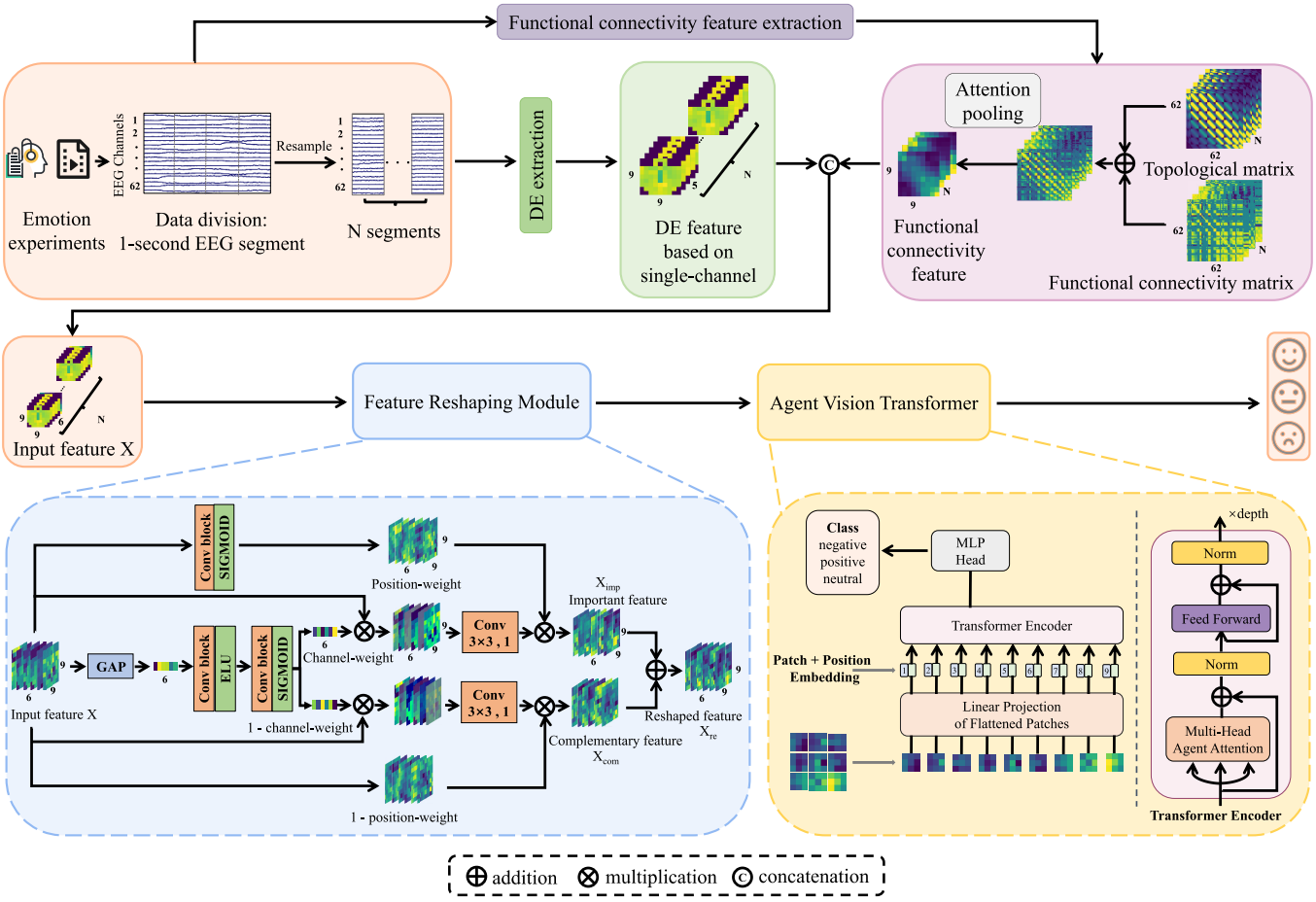


Fig. 1. Overall framework (STRAViT) of EEG-based emotion recognition: DE features and attention-enhanced connectivity matrices are extracted from multiband EEG signals; the FRM reshapes the features by enhancing important and complementary information; the AViT leverages a small set of learnable agent tokens to efficiently aggregate the reshaped features for emotion classification.

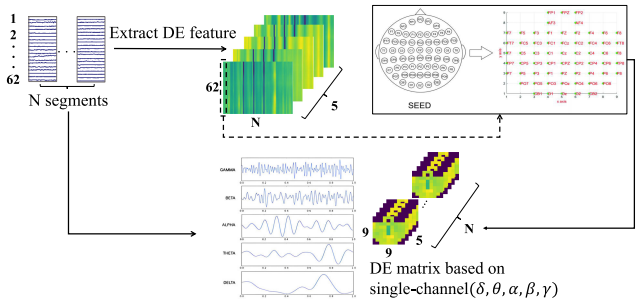


Fig. 2. 2-D DE matrices are constructed by first segmenting EEG signals into one-second epochs, followed by bandpass filtering into five frequency bands. DE features are then computed for each channel and each frequency band, and finally arranged according to the ESI NeuroScan 62-electrode system to form 9×9 matrices, with nonelectrode positions zero-padded.

function of x. Assuming that the EEG signals follow a Gaussian distribution $N(\mu, \sigma^2)$, the DE can then be simplified as follows:

$$DE = -\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) \cdot \log\left(\frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)\right) dx = \frac{1}{2}\log\left(2\pi e\sigma^2\right) \quad (2)$$

where μ and σ^2 denote the mean and variance of the EEG signal in the corresponding frequency band. Thus, for each EEG segment, we obtain a feature vector containing five frequency bands on each channel, which finally forms a 3-D feature tensor of size $5 \times N \times 62$, where five denotes the frequency bands, N represents the sample segments, and 62 indicates the EEG channels. Finally, the DE values are arranged into a 9×9 feature matrix based on the ESI NeuroScan 62-electrode system, with nonelectrode positions zero-filled to preserve spatial information, resulting in an $N \times 5 \times 9 \times 9$ DE feature matrix, as shown in Fig. 2.

2) Functional Connectivity Feature Extraction: Although DE features effectively portray the energy distribution in each frequency band, they ignore the functional connectivity between channels due to their inherently single-channel feature. DE characterizes the local signal dynamics of individual EEG channels by capturing time-domain variations and frequency-specific energy distributions. In contrast, functional connectivity features model the interactions and dependencies among different EEG channels. Since emotion-related neural processes involve coordinated activity across multiple brain regions, functional connectivity provides complementary information that cannot be captured by purely temporal or spectral features.

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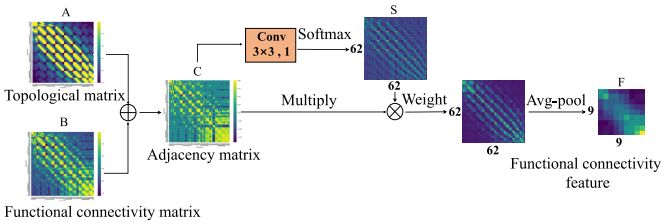


Fig. 3. Adjacency matrix is constructed by summing a topological matrix based on Euclidean distances and a functional connectivity matrix based on Pearson correlation. This matrix is then processed by an attention pooling structure, resulting in a low-dimensional representation referred to as the functional connectivity feature, which preserves key spatial patterns while reducing dimensionality.

There are several commonly used strategies to construct the matrix in EEG analysis, including functional connectivity [61], distance-based [62], and neural networks. EEG signals can be interpreted as the superposition of neural activities generated by spatially distributed dipoles [63], which implies that electrodes located far apart may still exhibit strong statistical dependencies. Consequently, matrices constructed solely based on Euclidean distance primarily model local spatial relationships and may overlook meaningful long-range functional correlations between distant electrodes. To address this limitation, we introduce functional connectivity feature and further incorporate spatial structural information into their representation, so as to construct spatially organized functional connectivity inputs for the model.

Specifically, a distance-based adjacency matrix derived from Euclidean distances is first constructed to encode the spatial topology of EEG channels, serving as a structural prior that preserves local neighborhood relationships. Meanwhile, a functional connectivity matrix based on Pearson correlation captures the global functional interactions among brain regions, as shown in Fig. 3. These two matrices are then combined by element-wise summation to produce a functional connectivity representation with embedded spatial information, which integrates global functional interactions with local spatial structure. The proposed method incorporates the following two components.

Topological Matrix: The distance-based adjacency matrix $A \in R^{62\times62}$ is built by computing Euclidean distances between EEG channels, capturing their local spatial dependencies. The formula for constructing A_{ij} is as follows:

$$A_{ij} = \min\left(1, \frac{\delta}{d_{ij}^2}\right) \tag{3}$$

where d_{ij} denotes the Euclidean distance between nodes i and j, and δ is the calibration constant, usually taken as 5. According to related studies [64], δ setting it to five represents the retention of 20% of connections, which improves the efficiency of the model. The formula enhances the strength of connections between spatially adjacent channels, thus emphasizing spatial localization.

Functional connectivity matrix: Using functional connectivity, the Pearson correlation was applied to measure the relationships between channel pairs, forming the global connectivity matrix $B \in \mathbb{R}^{62\times62}$. The similarity between channels is calculated as follows:

$$p_{xy} = \frac{cov(x,y)}{\delta_x\delta_y} = \frac{E\left[(x-\bar{x})(y-\bar{y})\right]}{\delta_x\delta_y} \tag{4}$$

where functions cov and E represent covariance and expectation, respectively, x and y refer to the values of the corresponding nodes, $\bar{x}$ and $\bar{y}$ are the mean values of the eigenvectors, and δ_x and δ_y denote their variances. The computation of B_{ij} is given by the following formula:

$$B_{ij} = \begin{cases} p_{ij}, & \text{if } |p_{ij}| > \lambda \\ 0, & \text{else} \end{cases}, \quad i,j = 1,2,\ldots,62 \tag{5}$$

where B_{ij} denotes the correlation coefficients between nodes i and j, and λ is set to the correlation coefficient threshold. Connections are preserved if the absolute correlation between nodes exceeds λ. The threshold is introduced to suppress weak correlations while retaining meaningful functional interactions. In this work, λ is set to 0.5 based on preliminary evaluations across several candidate values (e.g., 0.3, 0.4, 0.5, 0.6 and 0.7), which has been commonly adopted in prior EEG connectivity studies to achieve a balance between network sparsity and information preservation. Finally, the topological matrix and the functional connectivity matrix are added and fused to obtain the final adjacency matrix $C \in \mathbb{R}^{62\times62}$.

To prevent redundant information from affecting feature representation and to facilitate the fusion of DE and spatial connectivity matrices, we design a lightweight attention pooling structure. Rather than performing uniform spatial downsampling, this structure aggregates spatial information in a task-driven manner and compresses the original 62×62 connectivity matrix into a more compact 9×9 spatial representation, aiming to balance representational compactness and discriminative capability. Specifically, a 3×3 convolution is first applied to the adjacency matrix C, followed by a Softmax function that converts the convolution output into a probability distribution, generating an attention weight matrix. This attention matrix S is then element-wise multiplied by C to adaptively reweight the features. By selectively emphasizing informative regions while suppressing redundant responses, an average pooling operation is applied to obtain the spatial feature matrix $F \in \mathbb{R}^{9\times9}$. The formula for attention pooling is as follows:

$$F = \text{AvgPool}\left(\text{Softmax}\left(\text{Conv}_{3\times3}(C)\right) \odot C\right) \tag{6}$$

where $C \in \mathbb{R}^{62\times62}$ and $F \in \mathbb{R}^{9\times9}$. Through the above operations, spatial features of EEG channels are effectively extracted and processed.

B. Feature Reshaping Module

Effectively integrating spatial–temporal features remains a major challenge in EEG-based emotion recognition, particularly in cross-subject scenarios. Simply concatenating two features may lead to redundancy or semantic conflict, thereby limiting model performance. To address this issue, we propose FRM, which aims to refine and reorganize fused features rather than altering their spatial resolution or dimensionality.

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In this work, reshaping refers to adaptively reweighting and recomposing feature representations along channel and spatial dimensions to emphasize emotion-relevant patterns while suppressing redundant or subject-specific responses.

Specifically, FRM adopts a dual-branch attention structure consisting of a main attention branch and a reverse-attention complementary branch. The main branch highlights dominant feature responses that contribute most strongly to emotion discrimination, whereas the complementary branch explicitly models less-attended information that may be suppressed by primary attention. This complementary mechanism, inspired by reverse-attention strategies widely used in visual recognition, has been shown to enhance the completeness and robustness of learned representations by explicitly modeling the information that is often suppressed by the primary attention. By jointly considering salient and complementary patterns, FRM alleviates the risk of overfitting to subject-specific characteristics and promotes the learning of more generalizable emotion representations.

The $5 \times 9 \times 9$ DE feature matrix is concatenated with the $1 \times 9 \times 9$ functional connectivity matrix to form a fusion matrix $6 \times 9 \times 9$, which is used as the input $X \in \mathbb{R}^{6\times9\times9}$ to the FRM. The module first employs a channel attention branch and a positional attention branch to extract importance weights for each channel and each spatial location. The process is described as follows:

$$\begin{cases} X' = \delta\left(Conv_{1\times1}\left(GAP\left(X\right)\right)\right) \\ X_{ch} = \sigma\left(Conv_{1\times1}\left(X'\right)\right) \\ X_{loc} = \sigma\left(Conv_{3\times3}\left(X\right)\right) \end{cases} \tag{7}$$

where GAP stands for global average pooling, δ represents the ReLU activation, σ denotes the Sigmoid activation function, $X_{ch} \in \mathbb{R}^{6\times1\times1}$ for channel attention, and $X_{loc} \in \mathbb{R}^{6\times9\times9}$ for positional attention. Subsequently, a dual-branch structure is constructed as follows:

$$\begin{cases} X_{att} = X_{ch} \odot X \\ X_{1-att} = (1 - X_{ch}) \odot X \end{cases} \tag{8}$$

where $\odot$ denotes channel-wise multiplication, $X_{att} \in \mathbb{R}^{6\times9\times9}$ is the feature enhanced by channel attention, and $X_{1-att} \in \mathbb{R}^{6\times9\times9}$ is the complementary feature from inverse attention. The former branch enhances the prominent channels, while the latter supplements the potentially overlooked background information. Subsequently, a 3×3 convolution is performed on X_{att}, X_{1-att}. Finally, the output features of the two branches can be expressed as

$$\begin{cases} X_{imp} = \left(Conv_{3\times3}\left(X_{att}\right)\right) \otimes X_{loc} \\ X_{com} = \left(Conv_{3\times3}\left(X_{1-att}\right)\right) \otimes (1 - X_{loc}) \end{cases} \tag{9}$$

where $\otimes$ denotes element-wise multiplication, $X_{imp} \in \mathbb{R}^{6\times9\times9}$ is the feature enhanced by channel and position attention, and $X_{com} \in \mathbb{R}^{6\times9\times9}$ is the complementary feature. The former further emphasizes high-weight spatial regions based on the enhancement of salient channels, while the latter compensates for complementary features that may be overlooked by important features. Based on these attention weights, the main branch enhances salient feature responses, while the reverse branch recovers complementary information via inverse attention. Finally, the two branches are fused via element-wise addition to generate the reshaped EEG feature $X_{re} \in \mathbb{R}^{6\times9\times9}$.

Compared with direct feature concatenation, FRM establishes an effective balance between salient feature enhancement and background information preservation through its dual-branch structure. By guiding the model to learn more stable and representative feature representations, FRM improves robustness to subject variability and noise perturbations, which is beneficial for extracting generalizable emotion-related features. In addition, FRM preserves the same feature dimensionality as its input, allowing seamless integration with the backbone network. With only five lightweight convolution operations, it introduces minimal computational and parameter overhead, making it a plug-and-play module that can be easily incorporated into various deep architectures.

C. Agent ViT

Given the form of the input EEG fusion features $X_{re} \in \mathbb{R}^{B\times6\times9\times9}$ (B denotes the batch size), they can be regarded as multichannel images with a fixed spatial resolution, allowing image-like processing techniques to be applied for local and global feature extraction. Such a structured representation provides a natural foundation for introducing the ViT framework, enabling the model to capture global inter-regional dependencies while retaining local spatial structure. In contrast, traditional Transformer self-attention relies on full softmax-based pairwise interactions among all tokens, which, when applied to high-dimensional multipatch EEG features, may introduce excessive redundant interactions and adversely affect the learning of generalizable representations. Inspired by [23], we incorporate agent attention into the attention module of ViT, and apply it for the first time in EEG based emotion recognition. By introducing agent tokens to focus on local regions, this mechanism bridges the gap between global receptive fields and localized responses, thus improving the model's emotion classification performance.

First, the reshaped EEG feature X_{re} is split into nonoverlapping patches based on a predefined patch size, with the total number of patches denoted as $num_patches$. Each patch is then flattened and projected into a vector of dimension dim through a linear transformation, resulting in a sequence of embedded patch tokens. To preserve spatial information, learnable positional embeddings are added to these patch tokens. Finally, the position-enhanced patch embeddings $X_{input} \in \mathbb{R}^{B\times N\times C} (N = num_patches,\ C = dim)$ are used as the input sequence to the following Transformer layers.

Subsequently, the embedded tokens are passed into the multihead agent attention module, as shown in Fig. 4. Unlike conventional self-attention, agent attention does not directly compute pairwise similarities between all patch tokens; instead, it introduces agent tokens to aggregate and redistribute information in a structured manner. The generation of Query, Key, and Value is formulated as follows:

$$\begin{cases} Q = X_{input} W_Q \\ K = X_{input} W_K \\ V = X_{input} W_V \end{cases} \tag{10}$$

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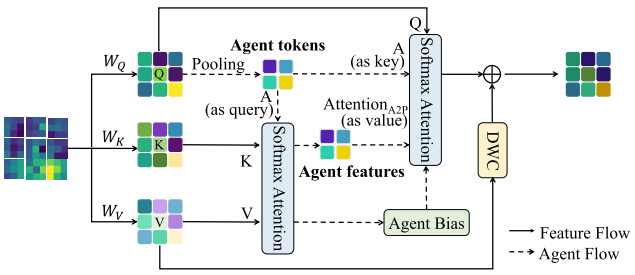


Fig. 4. Illustration of the agent attention module. Instead of using the traditional softmax mechanism, we introduce agent tokens to dynamically modulate attention and enhance feature representation.

where $W_Q, W_K, W_V \in \mathbb{R}^{C\times C}$ are learnable parameters and $Q, K, V \in \mathbb{R}^{B\times N\times C}$. To realize multihead attention, which captures the interaction of information in different subspaces, we reshape it as: $Q, K, V \in \mathbb{R}^{B\times h\times N\times d}(h = num_heads,\ d = C/num_heads)$. Different from traditional ViT, the agent attention mechanism incorporates an extra set of agent vectors A into the conventional attention triad (Q, K, V), forming an extended structure (Q, A, K, V). Agent tokens are generated from Q features by adaptive average pooling

$$A = AdaptiveAvgPool2d\,(Q,\ output_size = (r, r)) \tag{11}$$

where $r \times r = agent_num$ denotes the number of agent vectors, $A \in \mathbb{R}^{B\times agent_num\times C}$, and further reshaped as $A \in \mathbb{R}^{B\times h\times agent_num\times d}$.

In agent attention, pairwise similarities between Q and K are not directly computed. Instead, a small set of agent vectors A is introduced to aggregate information from K and V and subsequently pass it to Q. Specifically, A is first used as the query in an attention operation over K and V, aggregating information from all values to obtain the agent features Att_{A2P}. In the subsequent attention step, A is used as the key and Att_{A2P} as the value, with Q as the query. This allows the global information encoded in the agent features to be propagated back to each feature token, yielding the final output representation. This approach avoids explicit query-key similarity computation and instead enables information exchange through the agent vectors A. The corresponding formula is given as follows:

$$\begin{cases} Att_{A2P} = Softmax\left(\dfrac{AK^T}{\sqrt{d}} + B_{A2P}\right)V \\ Att_{P2A} = Softmax\left(\dfrac{QA^T}{\sqrt{d}} + B_{P2A}\right)Att_{A2P} \end{cases} \tag{12}$$

where B_{A2P}, B_{P2A} is the agent bias to allow different agent tokens to focus on different locations in the feature, $Att_{A2P} \in \mathbb{R}^{B\times h\times agent_num\times d}$, $Att_{P2A} \in \mathbb{R}^{B\times h\times N\times d}$. These steps enable the agent vectors to effectively aggregate and distribute global information while maintaining dynamic interactions with local patches.

To further improve the model's information perception, a deep convolution module is introduced to capture local features and add them to the output Att_{P2A}. Then, a linear projection and a dropout layer are applied to generate the final output O of the agent attention module. The process is defined as follows:

$$O = Dropout\,(Linear\,(Att_{P2A} + DWConv\,(V)))\,. \tag{13}$$

A residual connection followed by layer normalization is first applied, and the result is then passed through a feedforward (FF) network, which consists of two linear transformations and GELU. Finally, the output is fed into the MLP header to get the classification result

$$\begin{cases} X' = LayerNorm\left(X_{input} + O\right) \\ Output = MLP\left(LayerNorm\left(X' + FF\left(X'\right)\right)\right). \end{cases} \tag{14}$$

Introducing the agent attention mechanism not only enhances the model's capacity to capture spatial information but also strengthens its ability to model inter-regional dependencies by incorporating semantic interactions among local regions. Compared with traditional full softmax self-attention, agent attention offers a more efficient and targeted mechanism for integrating sparse local activations with global context, which better aligns with the intrinsic characteristics of EEG-based emotion recognition.

IV. Experiments and Results

In this section, a series of experiments are performed to demonstrate the reliability of the STRAViT network, all conducted on an NVIDIA Tesla V100 GPU. Before detailing the experimental settings, we first clarify the emotion classification task and data representation adopted in this work. The proposed STRAViT framework addresses a discrete emotion classification problem based on EEG signals. Specifically, EEG signals are segmented into fixed-length temporal windows, and each EEG segment is treated as an individual instance for emotion recognition. For data representation, each EEG instance is encoded using DE features extracted from five frequency bands and a spatially informed functional connectivity representation. The DE features are mapped onto a 2-d spatial grid to form a $5 \times 9 \times 9$ tensor, while the functional connectivity feature is projected into a $1 \times 9 \times 9$ spatial representation. These two components are concatenated along the channel dimension, resulting in a unified feature tensor of size $6 \times 9 \times 9$, which serves as the input to the proposed STRAViT model across all datasets and experimental settings.

A. Related Experiment Settings

1) Dataset Description: We evaluated our model using two widely used EEG emotion datasets.

1) *SEED [10]:* EEG recordings were recorded from 15 participants when they watched 15 film clips that evoked three emotion categories—positive, neutral, and negative. Each participant completed three sessions, each consisting of 15 trials (five trials per emotion), comprising a total of 45 trials per subject.
2) *SEED-IV [25]:* Following the same protocol, 15 participants viewed 24 clips across three sessions, with four target emotions: happy, neutral, sad, and fear. This yielded 72 EEG recordings per participant.

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In both datasets, EEG signals were collected using the ESI NeuroScan system with 62-channel EEG caps, uniformly downsampled to 200 Hz, ensuring consistency in subsequent feature extraction and analysis.

2) Experiment Protocol: To evaluate the effectiveness of the proposed model, we conduct both subject-dependent and subject-independent experiments on the SEED and SEED-IV datasets.

1) *Subject-Dependent Setting:* Training and testing are performed on different trials from the same subject. For SEED, each session includes 15 trials, with the first 9 used for training and the remaining 6 for testing. For SEED-IV, each session consists of 24 trials, where the first 16 are used for training and the last 8 for testing. Results are averaged across the three sessions and reported as "mean ± std."
2) *Subject-Independent Setting:* The leave-one-subject-out (LOSO) cross-validation scheme is utilized to evaluate subject-independent performance. In each round, one subject is held out for testing, while the remaining data are used for training. This process is repeated until every subject has been tested once. Final performance is reported as the average accuracy across all subjects, also in the form of "mean ± std."

3) Experiment Settings: In this work, these parameters are set to fixed values based on preliminary experiments and recommendations from prior studies. For the subject-dependent experiments, a batch size of 256 is used for both the SEED and SEED-IV datasets. The Adam optimizer is employed with an initial learning rate of 0.0001 and a weight decay of 0.0001. In the independent setting, batch sizes are set to 256 for the SEED dataset and 4096 for the SEED-IV dataset. The Adam optimizer is applied with an initial learning rate of 0.0001 and a weight decay parameter of 0.01. To enhance convergence speed and model generalization, the OneCycle learning rate policy is employed, where the learning rate gradually increases to 0.001 before decreasing throughout the training process. The dropout rate is set as 0.1, and the model is trained with cross-entropy loss. For EEG signal processing, the SEED dataset is segmented using one-second windows and resampled to 240 segments per trial. Under the subject-dependent setting, this results in 2160 training samples and 1440 testing samples per session, while under the subject-independent LOSO setting, the model is trained on 151 200 samples from 14 subjects and evaluated on 10 800 samples from the held-out subject. Similarly, the SEED-IV dataset is segmented using four-second windows and resampled to 60 segments per trial. This leads to 960 training samples and 480 testing samples in the subject-dependent setting, and 60 480 training samples with 4320 testing samples in the subject-independent setting. The input to the model is a $6 \times 9 \times 9$ feature matrix with a *patch_size* of 3. The Transformer depth is set to 1 to balance modeling capacity and generalization performance. Given the relatively limited training data and the already refined feature representations produced by the preceding modules, deeper Transformer stacks tend to introduce unnecessary model complexity and increase the risk of overfitting. A single-layer Transformer is therefore sufficient to capture essential global dependencies while maintaining stable and robust performance. In the agent attention module, h is set to 8 and *agent_num* is set to 9.

TABLE I
COMPARISON OF AVERAGE ACCURACIES (%) ACROSS DIFFERENT METHODS UNDER SUBJECT-DEPENDENT SETTINGS

Method	Years	Datasets	
		SEED	SEED-IV
SVM [17]	1999	84.0 ± 9.7	56.6 ± 20
DGCNN [31]	2018	90.4 ± 8.5	—
BiDANN [32]	2018	92.4 ± 7.0	—
BiHDM [13]	2020	93.1 ± 6.1	74.4 ± 14
4D-aNN [40]	2022	96.3 ± 1.9	79.8 ± 10
Conformer [48]	2022	95.3 ± —	—
GMSS [65]	2022	96.5 ± 4.6	86.4 ± 11
RFPN [33]	2023	96.8 ± —	91.6 ± —
AMDET [49]	2023	97.2 ± 0.93	87.3 ± 1.8
CIT-EmotionNet [50]	2024	98.6 ± 1.4	92.1 ± 5.3
DGAT [37]	2024	91.7 ± 7.8	—
MECAM [41]	2024	97.9 ± 2.4	94.8 ± 2.6
EEG-SWTNS [46]	2024	94.8 ± 7.2	79.5 ± 11
DSSTNet [39]	2025	95.2 ± 5.9	—
Ours	2025	**98.7 ± 3.0**	**95.5 ± 6.1**

"*Underline*" indicates the highest recognition accuracy among all compared methods.

TABLE II
COMPARISON OF AVERAGE ACCURACIES (%) ACROSS DIFFERENT METHODS UNDER SUBJECT-INDEPENDENT SETTINGS

Method	Years	Datasets	
		SEED	SEED-IV
SVM [17]	1999	56.7 ± 16	38.0 ± 13
DGCNN [31]	2018	80.0 ± 9.0	—
BiDANN [32]	2018	84.1 ± 6.9	—
BiHDM [13]	2020	85.4 ± 7.5	69.0 ± 8.7
CLISA [66]	2022	86.4 ± 6.4	—
MADA [55]	2023	86.2 ± 7.9	59.3 ± 14
DGAT [37]	2024	85.9 ± 7.5	—
JD-IRT [54]	2024	83.2 ± 11	52.1 ± 16
EEG-SWTNS [46]	2024	80.1 ± 11	66.7 ± 10
GCD-JFSE [57]	2025	78.7 ± 15	59.0 ± 11
DAEEGViT [52]	2025	79.2 ± 6.9	62.4 ± 10
Ours	2025	**87.2 ± 5.7**	**72.5 ± 6.6**

"*Underline*" indicates the highest recognition accuracy among all compared methods.

B. Results

1) Comparison With Other Models: To verify the constructed STRAViT network performance, we perform both subject-dependent and subject-independent experiments using the SEED and SEED-IV datasets, and compare them with advanced methods. Table I summarizes the recognition accuracy of various methods in subject-dependent experiments. Traditional machine learning approaches, such as SVM

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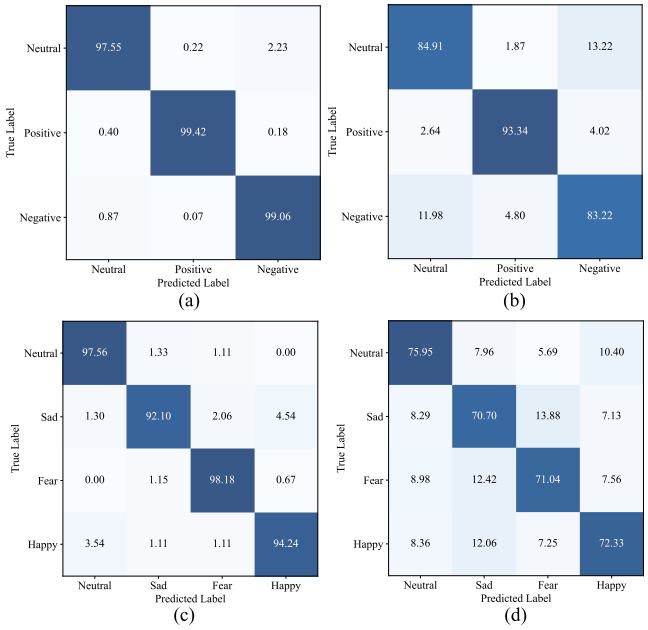


Fig. 5. (a) and (c) Subject-dependent confusion matrices for the SEED and SEED-IV datasets, whereas (b) and (d) subject-independent confusion matrices for the same datasets. The true and predicted classes are indicated on the X- and Y-axes, respectively.

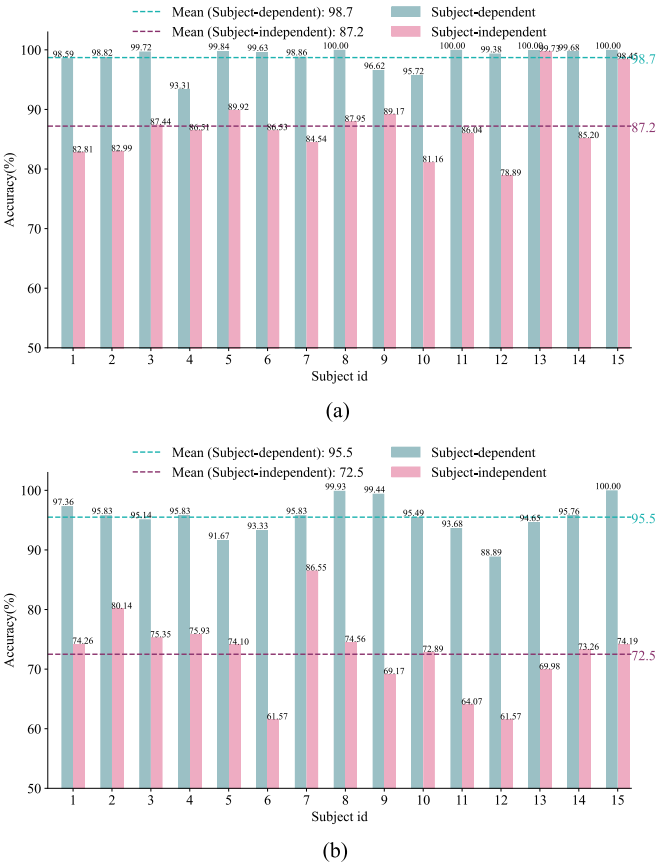


Fig. 6. Recognition performance for each subject. (a) and (b) Accuracy results on the SEED and SEED-IV datasets.

[17], show comparatively lower classification performance on both datasets, suggesting their insufficient capacity to model complex emotional patterns in EEG signals. In contrast, deep learning methods like BiDANN [32], BiHDM [13], CIT-EmotionNet [50], and MECAM [41] show significant improvements in accuracy. Among all methods, our proposed model attains the top average recognition accuracy of 98.7% on the SEED dataset and 95.5% on SEED-IV, surpassing the best competing methods by 0.1% and 0.7%, respectively. These results demonstrate the model's strong feature representation and classification capabilities in subject-dependent strategies. Table II summarizes the recognition accuracy of various methods in subject-independent strategies. Our model obtains an average accuracy of 87.2% on the SEED dataset and 72.5% on the SEED-IV dataset, outperforming the strongest baseline methods by 0.8% and 3.5%, respectively. These findings demonstrate that the proposed STRAViT network maintains robust generalization capability in cross-subject emotion recognition tasks. Nevertheless, the noticeable performance gap between the subject-dependent and subject-independent settings underscores the ongoing difficulty posed by individual differences in EEG-based emotion recognition. Although the performance improvements over some competing methods are relatively modest in certain cases, it is worth noting that the proposed model consistently achieves gains across both datasets and experimental settings, indicating its stability and effectiveness in handling EEG-based emotion recognition tasks.

To further understand the model's performance across different emotions, we present confusion matrices in Fig. 5. For the SEED dataset, it can be observed that the "Positive" emotions are the easiest to recognize, while the "Negative" and "Neutral" emotions are more prone to being misclassified with each other. From Fig. 5(a) and (b), the recognition accuracy for "Neutral" and "Negative" emotions drops by approximately 12% and 15%, respectively, whereas "Positive" emotions decrease by only 6%, suggesting better cross-subject transferability for positive emotion features. For the SEED-IV dataset, "Neutral" emotions consistently show high accuracy in both experiments from Fig. 5(c) and (d). This indicates that EEG activity patterns under neutral states exhibit relatively high consistency across subjects, making them easier for the model to capture and transfer. Notably, "Fear" achieves high accuracy in the subject-dependent experiment but shows a marked decline in the subject-independent experiment, often being confused with "Sad." This result suggests that "Fear" emotions share certain similarities with other negative emotions at the level of EEG responses, and that such similarities are further amplified under cross-subject conditions, thereby increasing the difficulty of emotion classification. Experimental results are basically consistent with the research [16], [29].

In addition, the recognition performance for each individual subject under the STRAViT network is presented in Fig. 6. It is evident that subject-dependent experiments consistently achieve higher accuracies, which demonstrates the model's ability to effectively capture EEG features. In contrast, subject-independent experiments show a noticeable performance decline. For instance, in the SEED, the accuracy

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TABLE III
AVERAGE ACCURACIES (%) IN ABLATION EXPERIMENTS OF SEED AND SEED-IV

Dataset	Method	Subject-dependent	Subject-independent
SEED	ViT	89.3 ± 12	83.2 ± 6.1
	w/o FRM	90.4 ± 11	83.4 ± 6.8
	w/o AViT	96.1 ± 5.8	84.7 ± 7.0
	Ours	**98.7 ± 3.0**	**87.2 ± 5.7**
SEED-IV	ViT	86.5 ± 11	64.5 ± 7.2
	w/o FRM	87.1 ± 12	64.7 ± 8.1
	w/o AViT	90.7 ± 9.6	69.1 ± 7.8
	Ours	**95.5 ± 6.1**	**72.5 ± 6.6**

"*w/o*" indicates the exclusion of the specified module.

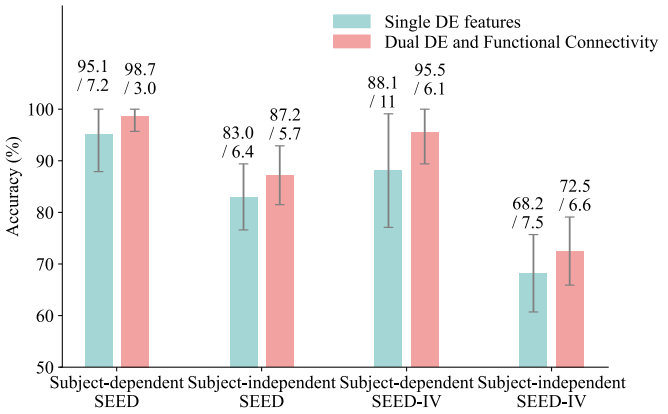


Fig. 7. Comparison of recognition performance between single-feature and dual-feature input models.

of subject 12 drops by 20.5%, and in the SEED-IV, the accuracy of subject 6 decreases by 31.8%. Additionally, the SEED dataset yields a higher mean accuracy compared to SEED-IV. This discrepancy may be attributed to the fact that the SEED-IV dataset contains a larger number of movie clips than the SEED dataset, which is likely to introduce increased intersignal variability or noise.

2) Ablation Study: To further analyze the contribution of each module in STRAViT, ablation studies were conducted on both datasets. The baseline is the ViT model without both modules. We tested the performance after removing each module individually under two types of experiments. The corresponding results are shown in Table III.

We could observe from Table III that the baseline model ViT achieves the lowest emotion recognition accuracy when the FRM and AViT are removed simultaneously. Incorporating either of the two modules into the STRAViT network improves the recognition performance, while integrating both modules simultaneously leads to a further accuracy improvement. In addition, introducing the FRM module significantly improves recognition accuracy: on subject-dependent experiments, SEED and SEED-IV accuracy increased by 6.8% and 4.2%, respectively; for subject-independent experiments, the improvements were 1.5% and 4.6%. Similarly, adding the AViT module also enhances recognition accuracy: improvements of 1.1% (SEED) and 0.6% (SEED-IV) in the subject-dependent experiments, and 0.2% (SEED) and 0.2% (SEED-IV) in the subject-independent experiments. These observations confirm that the STRAViT network benefits from the combined strengths of both modules, enhancing EEG-based emotion recognition. Therefore, the FRM and AViT modules are practical and effective components of the proposed approach.

To better highlight the advantage of dual-feature input over single-feature input, Fig. 7 presents the EEG emotion recognition accuracies achieved using both strategies on the SEED and SEED-IV datasets. It can be observed that the proposed STRAViT model achieves a relatively high recognition accuracy when using only DE features. Nevertheless, the performance is further enhanced when both DE features and functional connectivity features are simultaneously fed into the STRAViT model, leading to more competitive classification outcomes.

3) Contributions of DE in Different Frequency Bands: This subsection analyzes the DE features across five frequency bands for both datasets, as summarized in Table IV, revealing that each band contributes differently to emotion classification. Among them, the β band consistently delivers the highest accuracy across both experimental settings. This observation is consistent with existing neurophysiological findings that higher-frequency EEG rhythms are closely associated with emotional arousal, cognitive engagement, and affective processing [31], [60], [67]. Compared with lower-frequency bands, high-frequency rhythms are more sensitive to rapid neural responses and task-induced cortical activation. Consequently, DE features extracted from higher-frequency bands tend to exhibit stronger discriminative capability for emotion classification. Although individual band performance declines in subject-independent experiments, combining all frequency bands helps improve overall emotion recognition performance.

V. DISCUSSION

A. How Well Does STRAViT Generalize Across Subjects While Preserving Emotion Discriminability?

To clearly illustrate the benefits of our STRAViT approach in feature distribution, we chose representative subjects from the SEED and SEED-IV datasets and visualized the results of subject-independent experiments. The sample distributions were mapped using t-SNE, a technique for visualizing high-dimensional features. As shown in Fig. 8, black points denote the target domain data, while other colors indicate different source domain subjects. Specifically, Subject 5 from the SEED dataset and Subject 7 from SEED-IV were selected as the target domains. From (a) and (c), it can be clearly observed that the colored source domains are scattered and mostly overlapped with the black target domain, indicating a high degree of adaptability from source to target. This demonstrates the strong cross-domain generalization capability of our model and its effective performance in subject-independent experiments. From (b) and (d), it shows that different emotional

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TABLE IV
CLASSIFICATION RESULTS (%) USING DE FEATURES WITH SEPARATE AND ALL FREQUENCY BANDS ON THE SEED AND SEED-IV DATASETS

Frequency band	SEED		SEED-IV	
	Subject-dependent	Subject-independent	Subject-dependent	Subject-independent
Delta	68.7 ± 13	58.8 ± 6.1	70.3 ± 14	54.7 ± 7.9
Theta	78.1 ± 9.6	64.3 ± 9.6	71.9 ± 15	51.6 ± 7.0
Alpha	82.1 ± 12	58.6 ± 9.3	76.2 ± 14	50.7 ± 9.4
Beta	<u>88.2 ± 9.3</u>	<u>71.5 ± 9.0</u>	<u>79.7 ± 14</u>	<u>54.9 ± 8.4</u>
Gamma	85.6 ± 9.9	69.8 ± 8.2	77.6 ± 15	47.3 ± 9.8
All	**95.1 ± 7.2**	**83.0 ± 6.4**	**88.1 ± 11**	**68.2 ± 7.5**

"*Underline*" denotes the best recognition accuracy among separate frequency bands.

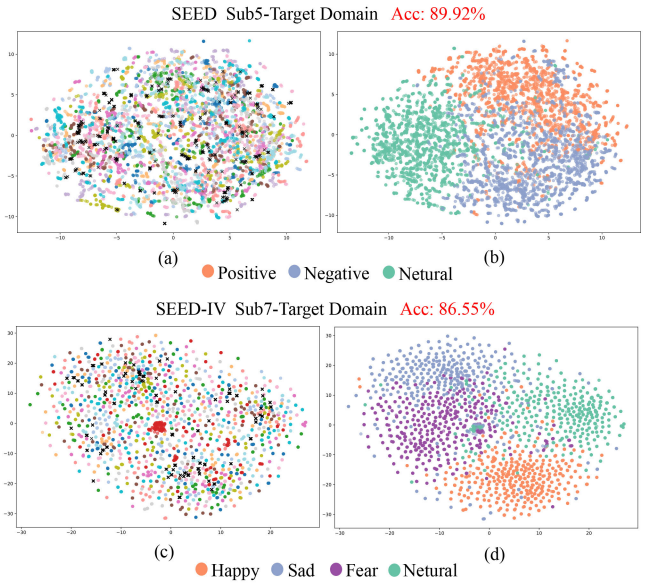


Fig. 8. Feature visualizations for selected subjects from the SEED and SEED-IV datasets are presented. (a) and (c) Feature distributions, with black points representing target domain data and other colors indicating various subjects in the source domain. (b) and (d) Distribution of emotion samples, where each color corresponds to a different emotion label. The classification accuracy of the STRAViT network is shown in red.

categories exhibit clear clustering trends in the feature distributions. This further confirms that our method effectively aligns domains while preserving discriminatory emotion information. Consequently, it enhances the reliability and stability of cross-subject emotion recognition.

B. How Does the FRM Capture Neurophysiologically Consistent Patterns Across Subjects and Emotions?

In order to further investigate the interpretability and neurophysiological reliability of the proposed model for cross-subject emotion recognition, we visualized the features extracted by the FRM module, showing the mean topographic maps of 15 subjects under different emotional conditions and five frequency bands. From the overall results, distinct spatial distribution patterns can be observed across multiple frequency bands for different emotional states, indicating that different functional brain regions play differentiated roles in emotional processing. In particular, the frontal, temporal, and parietal regions exhibit more prominent emotion-related information distributions, which is consistent with existing findings [68], [69]. These results suggest that the FRM module is capable of extracting stable and neurophysiologically meaningful discriminative patterns under cross-subject conditions, rather than relying on subject-specific local responses. Such observations provide intuitive neuro-level evidence supporting the model's favorable performance in subject-independent experiments.

As shown in Fig. 9, "Positive" emotions exhibit greater specificity across multiple bands, with particularly strong activations in the α, β, and γ bands. These patterns are more prominent in the left temporal and parietal regions. This observation is highly consistent with prior studies reporting that "Positive" emotions state exhibit stronger energy concentration and higher discriminability in high-frequency EEG bands. Existing research has widely indicated that the β and γ bands are closely associated with positive emotion processing and demonstrate relatively good stability and consistency across subjects [60]. In contrast, "Negative" emotions show relatively weaker activation in the β and γ bands, with activity mainly concentrated in the frontal area. Such response patterns exhibit greater intersubject variability, making them more prone to confusion with other emotional categories under cross-subject conditions. Similar phenomena have been reported in multiple EEG-based emotion studies, where the neural representations of "Negative" emotions are more strongly influenced by individual differences, resulting in relatively lower consistency of their spatial activation patterns at the group level [70]. "Neutral" emotions exhibit relatively balanced activation patterns across different frequency bands. This property leads to certain similarities in spatial distributions between "Neutral" and "Negative" emotions, making neutral states more susceptible to confusion under cross-subject conditions. These findings are consistent with previous studies reporting that neural responses to "Neutral" emotions are less concentrated and exhibit relatively weaker discriminability [67], [71].

Our FRM module captures activation patterns in these regions and frequency bands, demonstrating that the learned features reflect well-established emotion-related neural substrates. These results demonstrate that the proposed STRAViT network maintains good stability and generalization in cross-subject scenarios. Despite variations in emotional responses

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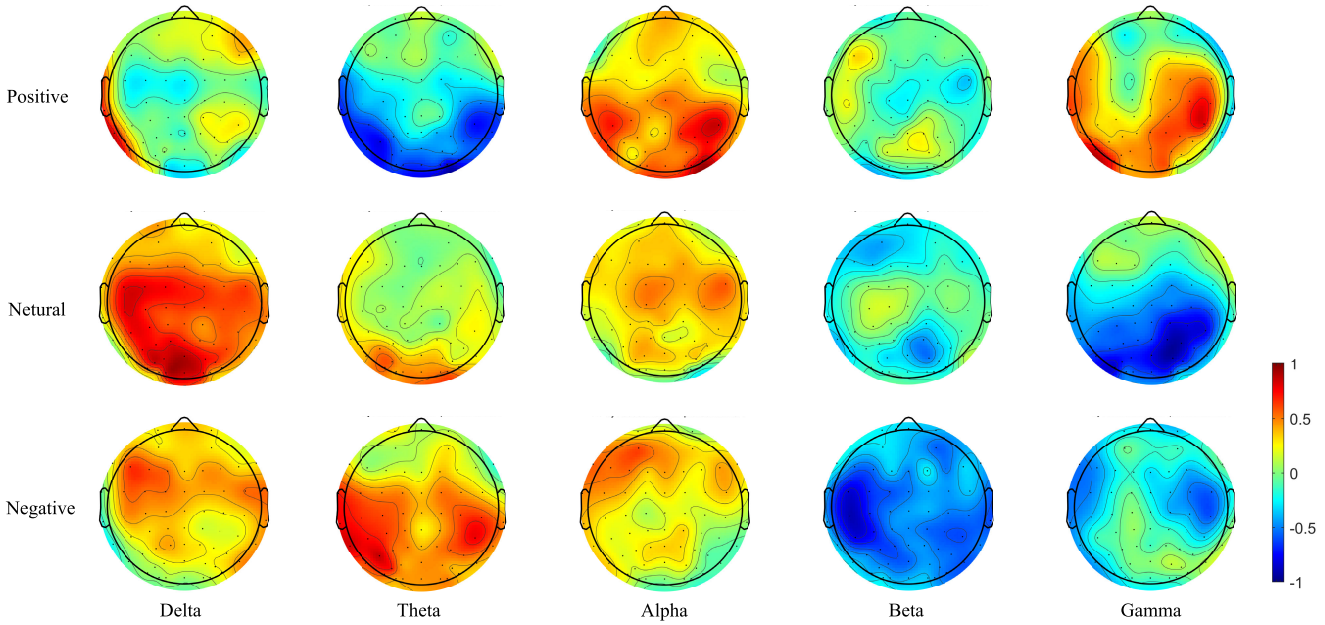


Fig. 9. Average brain topographic maps of 15 subjects across three emotional states and five frequency bands for the SEED dataset after FRM processing.

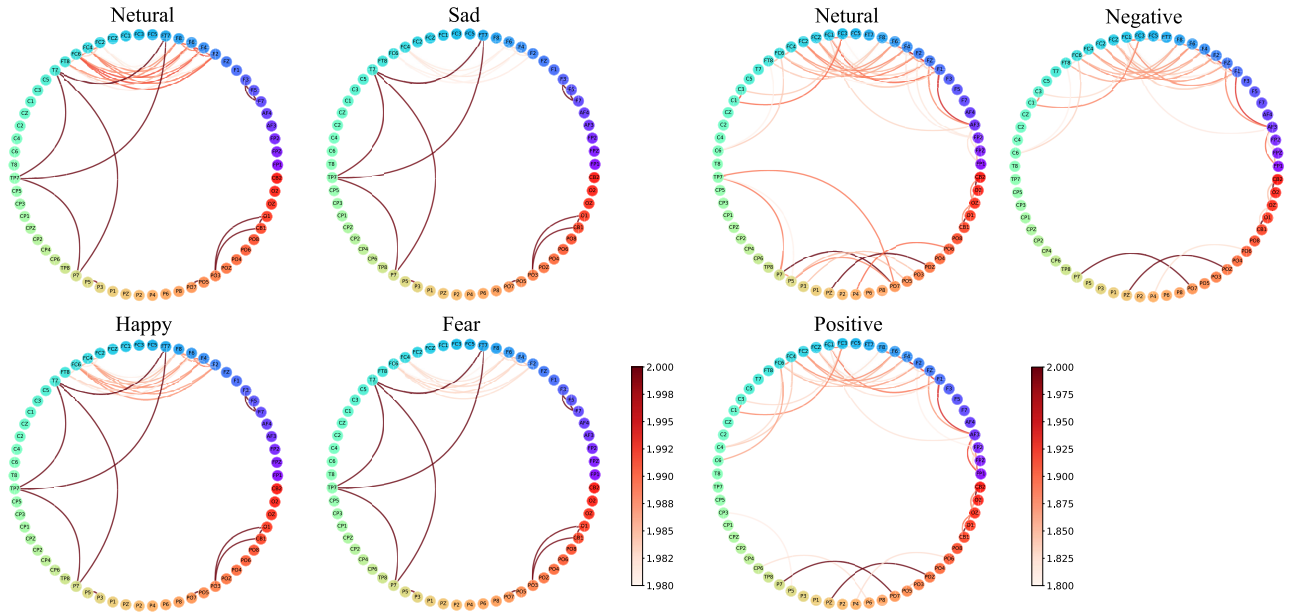


Fig. 10. Functional connectivity matrices for a representative subject from each dataset in the subject-dependent experiment are visualized. The curves linking the circles indicate the connectivity between different EEG channels, with darker curves representing stronger connection strengths.

across frequency bands, the model effectively captures discriminative spatial patterns, highlighting the FRM module's ability to model emotion-related brain regions and key frequency components. This further confirms that the model preserves neurophysiological consistency while ensuring robust emotion feature extraction and cross-subject reliability.

C. Contribution of EEG Functional Connectivity to Emotion Recognition

We visualized the functional connectivity patterns of a representative subject from each dataset in the subject-dependent setting. We observed similar patterns in the majority of subjects. For each emotion category, the corresponding connectivity matrices were averaged. Fig. 10 illustrates the distribution of interchannel connections, where circles represent EEG electrodes and curves indicate the connectivity of different EEG channels. The color intensity of the curves reflects connection strength, and only the strongest connections are displayed to better explore the correlation between connectivity matrices and labels. In the SEED dataset, the main connections are predominantly located in the frontal and parietal areas, which is consistent with prior research [72]. Notable connections include AF3–F1, P7–PO7, PZ–POZ, O1–CB1, and O2–CB2. In the SEED-IV dataset, significant connectivity is mainly found in the frontal, left temporal, and

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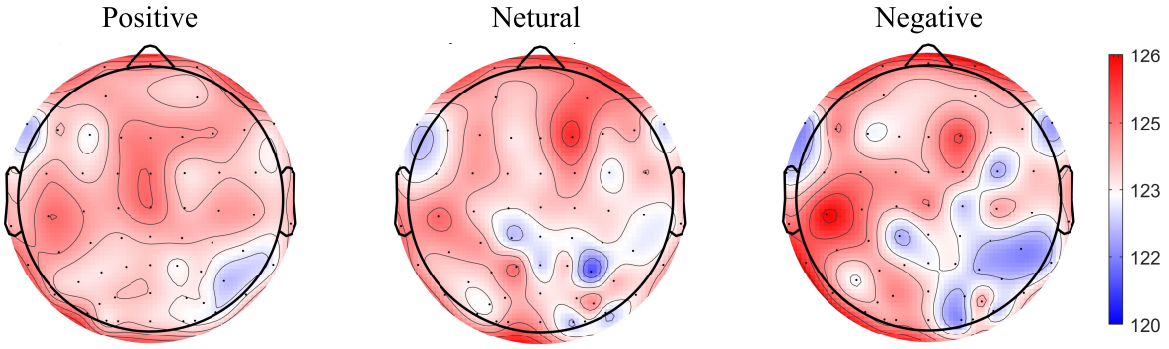


Fig. 11. Visualization of agent attention across three emotions for one subject. The attention maps were generated by averaging the outputs across all attention heads and agents, highlighting the model's channel preferences under different emotions. Warmer colors indicate higher attention values.

occipital areas, especially across connections such as T7–FT7, T7–TP7, FT7–TP7, T7–P7, TP7–P7, P3–P5, and PO3–O1. The major differences among the four emotion types appear in the frontal connections, where "Neutral" and "Happy" emotions show stronger links compared to "Sad" and "Fear" emotions.

D. Visualizing Agent Attention to Enhance the Interpretability of AViT in EEG-Based Emotion Recognition

To further investigate what the AViT module focuses on and to identify important EEG channels, we conducted a visualization analysis of the agent attention, as shown in Fig. 11. Since the spatial connectivity matrices do not directly correspond to specific electrode locations, we focused solely on the attention responses of the DE channels. When the model uses a *patch_size* of 1, the resulting attention maps can accurately reflect the model's channel pReferences. This modification is used solely for visualization purposes and does not affect the training or evaluation results of the proposed model.

We averaged the outputs across all attention heads and agents and visualized the attention heatmaps for three emotions from a representative subject. The results show that the model consistently attends to a subset of specific EEG channels across different emotion categories. In particular, channels such as F2, FP1, FP2, TP7, O1, Oz, and Cz exhibit relatively high attention responses across multiple emotional states. This observation indicates that the agent attention mechanism is capable of persistently focusing on core brain regions, including the frontal, temporal, and parietal regions, that are critical for emotion discrimination under varying emotional conditions, thereby demonstrating a certain degree of spatial consistency and stability, which is in line with previous findings [73], [74], [75]. Meanwhile, the attention distributions also exhibit noticeable differences across emotion categories. For instance, channels including F8, CP1, and FC4 show stronger attention responses under "Positive," whereas their responses are relatively attenuated under "Negative," suggesting that the model possesses discriminative capability at the emotion-category level. In addition, the overall attention patterns of neutral emotions are more similar to those of "Negative" emotions, while exhibiting more pronounced differences compared with "Positive" emotions. This phenomenon is consistent with the emotion separability trends observed in the confusion matrix analysis and the cross-subject recognition results discussed earlier.

This suggests that AViT not only effectively models global dependencies during training but also exhibits clear and neurophysiologically meaningful spatial attention patterns. Through visualization of attention responses, the model is shown to maintain both stability and category-level discriminability in its focus on key brain regions across different emotional states, thereby further validating the effectiveness and rationality of the AViT module for EEG-based emotion recognition from an interpretability perspective.

VI. Conclusion

This research proposes a STRAViT network for EEG-based emotion recognition. By combining DE and functional connectivity matrices through the FRM, the model effectively captures both localized temporal-frequency features and global spatial dependencies. The introduction of AViT further enhances the model's capacity to extract meaningful representations by leveraging learnable agent tokens. Experimental evaluations on the SEED and SEED-IV datasets confirm the feasibility and strong performance of the STRAViT in both subject-dependent and subject-independent settings. Additionally, ablation and visualization studies further demonstrate the model's effectiveness, stability, and interoperability.

Despite these strengths, a notable performance gap remains between subject-dependent and subject-independent settings, highlighting the challenges posed by intersubject variability in EEG signals. Although the FRM, and AViT modules of STRAViT contribute to capturing stable and discriminative features, residual differences among individuals in brain anatomy, electrode placement, and emotional responses continue to limit cross-subject generalization. Furthermore, although the proposed method achieves consistent improvements across different datasets and experimental settings, the overall accuracy gains remain relatively modest, indicating room for improvement in learning more discriminative and generalizable EEG features. Future work will focus on developing more effective feature representation strategies to achieve greater performance gains.

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In future work, we will further explore strategies to mitigate intersubject variability, including multimodal data integration and the investigation of more effective functional connectivity representations, particularly under subject-independent conditions. In addition, we will investigate its adaptation for real-time and edge-device deployment. This may involve simplifying or sparsifying functional connectivity computation, and applying standard model compression techniques such as quantization, pruning, or knowledge distillation to achieve a favorable tradeoff between recognition performance and computational efficiency.

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